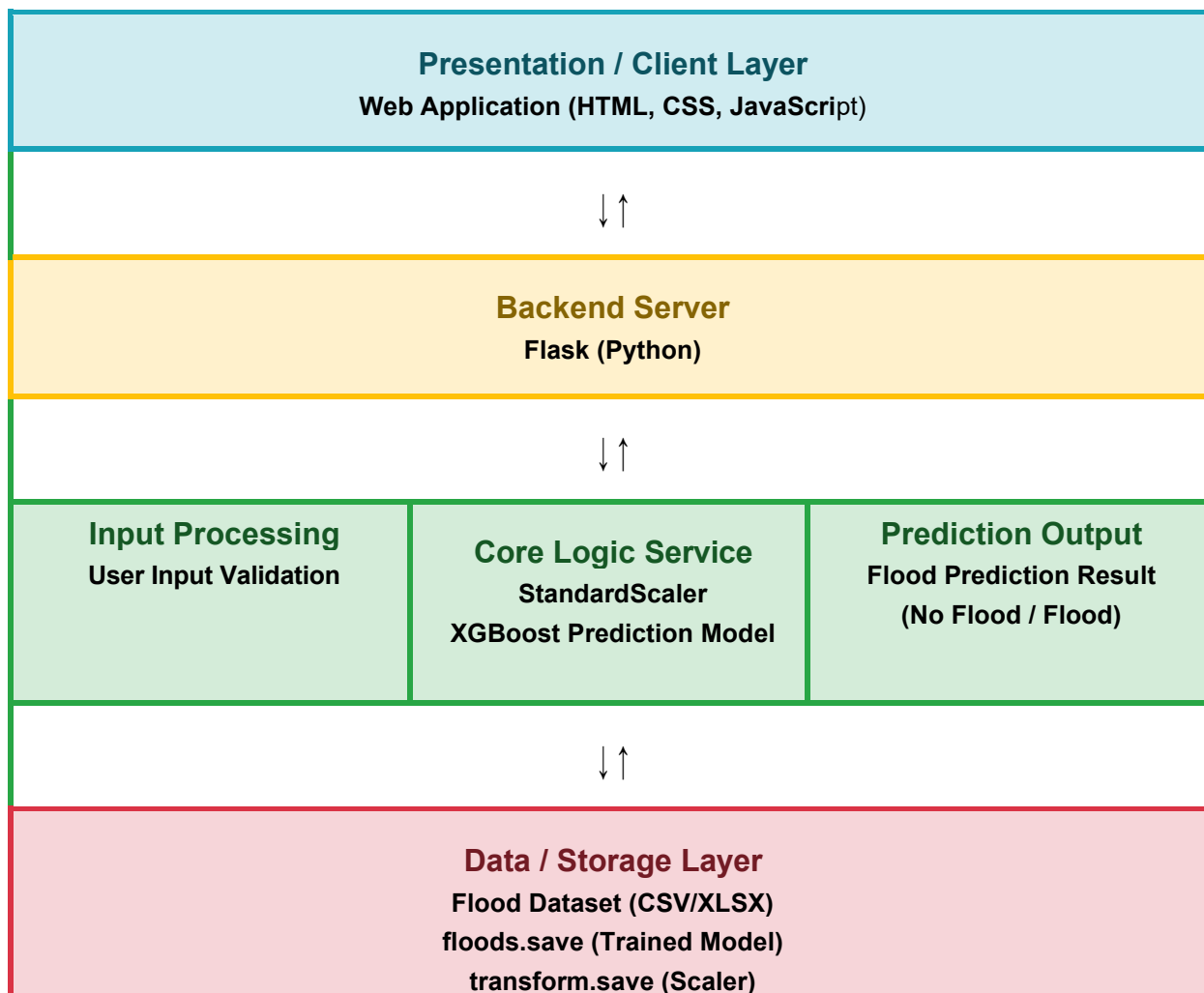


Solution Architecture

Date	1 July 2026
Team ID	XXXXXX
Project Name	Flood Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users enter weather parameters and view flood prediction results through web pages.	HTML, CSS, JavaScript
API Gateway / Backend Server	Receives user requests, validates input, communicates with the machine learning model, and returns prediction results to the web application.	Python, Flask
Core Logic Service (Machine Learning)	Preprocesses the input using the saved scaler and predicts flood occurrence using the trained XGBoost model.	Scikit-learn, XGBoost, Joblib
Data / Storage Layer	Stores the historical flood dataset, trained machine learning model, and scaler required for prediction.	CSV/XLSX Dataset, floods.save, transform.save